

Signal resampling

Write a Matlab function **SignalResample** that takes as input a [1 x N] signal **x** and sampling frequency **xFs**, as well as a resampling frequency interpolation with `interp1` to create and output signal **y**, a resampled version of **x** with sampling rate **yFs**. The steps are as follows:

- Find the total time duration **T** for the signal **x** with sampling frequency **xFs**
- Using **T**, define a time vector **tx**, which is the same size as **x**. Each entry in **tx** should contain the time for that entry, starting at time 0.
- Using **T**, define the new time vector, **ty** for our resampled time. The entries in **ty** should range from 0, in steps of $1/yFs$, to **T**.
- Use `'interp1'` to create **y** by resampling **x** to the new sampling rate using linear interpolation.

For example, if

```
x = [1:0.5:5.5];
xFs = 2;
yFs = 4;
```

the resampled x-values will be [1 1.25 1.5 1.75 2 2.25 2.5 2.75 3 3.25 3.5 3.75 4 4.25 4.5 4.75 5 5.25 5.5]

And if

```
x = [1:0.5:5.5];
xFs = 2;
yFs = 1;
```

the resampled x-values will be [1 2 3 4 5]

Function ?

Save

Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function y = SignalResample(x, xFs, yFs)
2
3 % Find the total time duration 'T'
4 T = (length(x)-1) / xFs;
5
6 % Using 'T', define a time vector 'tx'
7 tx = [0:1/xFs:T];
8
9 % Using 'T', define the new time vector, 'ty' for our resampled time.
10 ty = [0:1/yFs:T];
11
12 % Use interp1 to create y by resampling x
13 y = interp1(tx, x, ty, 'linear');
14
15 end
16
```

Code to call your function ?

Reset

```
1 x = [1:0.5:5.5];
2 xFs = 2;
3 yFs = 4;
4 y = SignalResample(x, xFs, yFs)
5
6 x = [1:0.5:5.5];
7 xFs = 2;
8 yFs = 1;
9 y = SignalResample(x, xFs, yFs)
```

Run Function



Assessment: All Tests Passed

Submit 

 Is output y correct?

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